

Notes: Dahlquist & Ibert (RFS, 2024)

Equity Return Expectations and Portfolios: Evidence from Large Asset Managers

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Capital market assumptions	2
2.2	Equity premium expectations	3
2.3	Objective expected return benchmarks	3
2.4	Allocation fund portfolios	3
3	Models and empirical specifications	4
3.1	Valuation cyclicity of expectations	4
3.2	Subjective and objective expected returns	4
3.3	Subjective term premium	4
3.4	Portfolio shares on expectations	4
3.5	Mean-variance benchmark	4
4	Identification and empirical design	5
4.1	What identifies countercyclicality	5
4.2	What identifies portfolio pass-through	5
4.3	Why mandates matter	5
4.4	Limits of the design	5
5	Tables and figures	6
5.1	Table 1 and Appendix sample evidence	6
5.2	Table 2, Figure 1, and Figure 2: equity premium expectations and CAPE	7
5.3	Table 3: subjective and objective equity premium expectations	8
5.4	Figure 3 and Table 4: subjective term premiums	9
5.5	Table 5, Figure 4, and Table 6: portfolio shares and U.S. equity expectations	9
5.6	Tables 7–9: tactical funds, aggregate equity shares, and bonds	11
5.7	Figure 5 and Tables 10–11: CFOs and professional forecasters	12
6	Short summary	14
7	Conclusion	14
7.1	Core conclusion	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** How do large asset managers form equity return expectations, and do these expectations show up in their portfolios?
- **Main contrast with retail investors.** Retail investors' return expectations are often procyclical and extrapolative. Large asset managers' U.S. equity premium expectations are countercyclical: high when valuations are low and low when valuations are high.
- **Data innovation.** The authors collect publicly reported capital market assumptions from large asset managers and match them to the managers' mutual fund portfolio allocations.
- **Expectation result.** Subjective U.S. equity premium expectations are strongly negatively related to the log CAPE. A higher CAPE predicts lower expected equity premiums.
- **Objective benchmark.** Asset managers' expectations line up with objective equity premium measures built from valuation-based forecasts, return-predictability regressions, option-implied equity premiums, and term premium models.
- **Portfolio result.** Asset managers with higher U.S. equity premium expectations allocate more to U.S. equities in allocation funds.
- **Magnitude result.** The portfolio response is statistically significant and larger than similar estimates for retail households, but still much smaller than the response predicted by a frictionless mean-variance portfolio choice model.
- **Mandate channel.** Investment mandates mute the pass-through from expectations to portfolio shares. Including fund fixed effects reduces the sensitivity because fund mandates absorb persistent differences in risk tolerance and target allocation.
- **Tactical funds.** Funds classified as tactical are more responsive to expectations, especially in log-share specifications.
- **Bond result.** Term premium expectations are more strongly related to fixed-income duration and maturity than to overall bond shares.
- **Comparison with other forecasters.** CFOs and professional forecasters differ from asset managers. Their short-horizon expectations often look procyclical in nominal-return terms, whereas asset managers' long-run equity premium expectations are countercyclical.
- **Economic takeaway.** Institutional investors' subjective expectations are not redundant: they vary across managers and map into portfolio demand, but institutional constraints and mandates dampen the response.

2 Data and measurement

2.1 Capital market assumptions

- The expectation data come from large asset managers' publicly reported capital market assumptions.
- These assumptions give forecasts for multiple asset classes, including U.S. equities, non-U.S. developed equities, emerging market equities, Treasury bills or cash, and fixed income.
- Forecast horizons vary by manager and by time. The paper works with both all horizons and the forecasts closest to ten years.
- The final sample includes hundreds of expectation observations and thousands of matched fund-quarter observations.

Interpretation: Capital market assumptions are not survey responses from households. They are public, reputation-sensitive forecasts produced by professional asset managers.

2.2 Equity premium expectations

The main subjective U.S. equity premium measure is an expected U.S. equity return minus an expected safe return:

$$EP_{m,t,h}^{US} = E_{m,t}[R_{t,t+h}^{US}] - E_{m,t}[R_{t,t+h}^{safe}], \quad (1)$$

where m is the asset manager, t is the forecast date, and h is the forecast horizon.

The paper uses several variants:

- U.S. equity premium over Treasury yield;
- U.S. equity return in nominal levels;
- U.S. equity premium over cash;
- non-U.S. developed and emerging market equity premium expectations.

Interpretation: The equity premium version is the cleanest measure for asset pricing because it removes some variation in the risk-free rate.

2.3 Objective expected return benchmarks

The paper compares subjective expectations with objective proxies:

- valuation-based expected returns using the CAPE;
- regression-based equity premium forecasts from realized excess returns and valuations;
- option-implied equity premium estimates;
- model-based term premium estimates for bonds.

Interpretation: This comparison asks whether asset managers' forecasts are behavioral extrapolation or professional forecasts that resemble standard objective valuation-based measures.

2.4 Allocation fund portfolios

- The portfolio tests focus on allocation funds managed by the same asset managers that report capital market assumptions.
- The main portfolio shares are U.S. equity, non-U.S. equity, bonds, cash, and other assets.
- The paper also studies aggregate equity shares, bond shares, duration, and maturity.
- Fund fixed effects are used to absorb stable mandates and strategic allocations.

Interpretation: Allocation funds are useful because they have some flexibility across asset classes while still being observable in regulatory mutual fund data.

3 Models and empirical specifications

3.1 Valuation cyclicalty of expectations

The main expectation regression is:

$$EP_{m,t,h} = \alpha_{m,h} + \beta \log(CAPE_t) + \gamma' X_t + \varepsilon_{m,t,h}, \quad (2)$$

where $\alpha_{m,h}$ are manager-by-horizon fixed effects and X_t can include past returns and the risk-free rate.

Interpretation: A negative β means expectations are countercyclical: asset managers expect higher equity premiums when valuations are low.

3.2 Subjective and objective expected returns

The paper estimates:

$$EP_{m,t,h}^{subj} = \alpha_{m,h} + \theta EP_{t,h}^{obj} + u_{m,t,h}. \quad (3)$$

Interpretation: A positive θ means asset managers' expectations move with objective valuation-based or model-based expected returns.

3.3 Subjective term premium

For an n -period bond, the subjective term premium is defined as:

$$rpy_t^{(n)} = y_t^{(n)} - \frac{1}{n} E_t \left[y_t^{(1)} + y_{t+1}^{(1)} + \cdots + y_{t+n-1}^{(1)} \right]. \quad (4)$$

Interpretation: The asset manager's expected return on cash gives the expected path of short rates, so the paper can back out subjective term premium expectations.

3.4 Portfolio shares on expectations

The main portfolio regression is:

$$w_{f,t}^{US} = \alpha + \beta EP_{m,t}^{US} + \Gamma Z_{f,t} + \mu_t + \varepsilon_{f,t}, \quad (5)$$

where $w_{f,t}^{US}$ is the U.S. equity share of allocation fund f managed by manager m .

The log specification is:

$$\log(w_{f,t}^{US}) = \alpha + \beta EP_{m,t}^{US} + \Gamma Z_{f,t} + \mu_t + \varepsilon_{f,t}. \quad (6)$$

Some specifications add fund fixed effects:

$$w_{f,t}^{US} = \alpha_f + \beta EP_{m,t}^{US} + \mu_t + \varepsilon_{f,t}. \quad (7)$$

Interpretation: Without fund fixed effects, the coefficient reflects both strategic mandates and tactical responses. With fund fixed effects, it is closer to within-fund time-series adjustment.

3.5 Mean-variance benchmark

A frictionless mean-variance investor would choose approximately:

$$w = \frac{1}{\gamma} \frac{\mu}{\sigma^2}, \quad (8)$$

so the sensitivity of portfolio weight to expected return is:

$$\frac{\partial w}{\partial \mu} = \frac{1}{\gamma \sigma^2}. \quad (9)$$

Interpretation: The empirical coefficients are much smaller than this benchmark, suggesting mandates, constraints, multiple assets, risk budgets, or organizational frictions attenuate the pass-through from beliefs to portfolios.

4 Identification and empirical design

4.1 What identifies countercyclicality

- Time-series variation in CAPE and other valuation measures is matched to asset managers' expectation publications.
- Manager-by-horizon fixed effects absorb stable differences in forecasting levels across managers and horizons.
- The coefficient on log CAPE captures whether a given manager's expectations are high or low relative to its own average when valuations change.

Interpretation: This is not a causal estimate of CAPE changing beliefs. It is a disciplined description of whether professional expectations move with valuations in the direction predicted by asset pricing models.

4.2 What identifies portfolio pass-through

- Asset managers differ in their equity premium expectations at a given date.
- Allocation funds differ in their U.S. equity shares.
- Year-month fixed effects absorb common market and macro conditions.
- Fund fixed effects absorb stable mandates and strategic allocations.

Interpretation: The strongest specifications ask whether the same fund changes its equity share when its manager's expectations change.

4.3 Why mandates matter

- Allocation funds are not unconstrained mean-variance portfolios.
- Target allocations, benchmark constraints, risk budgets, and tactical versus strategic mandates all affect portfolio flexibility.
- Fund fixed effects can therefore absorb part of the economically meaningful response from expectations to portfolios.

Interpretation: The attenuation relative to theory is not necessarily irrationality; it may reflect the institutional structure of delegated asset management.

4.4 Limits of the design

- Expectations are publicly reported forecasts, not necessarily the private beliefs of every portfolio manager.
- The sample focuses on large managers willing to publish capital market assumptions.
- Portfolio holdings are observed for mutual funds, not all mandates or separate accounts.
- Fund mandates may be both a mediator and a confounder.

Interpretation: The paper provides unusually direct evidence on institutional expectations and portfolios, but the estimates should be interpreted as portfolio pass-through within an institutional setting.

5 Tables and figures

The visuals below are directly cropped from the paper and grouped compactly.

5.1 Table 1 and Appendix sample evidence

Read:

- U.S. equity premium expectations over yield average about 3.02% across all horizons and 3.72% at horizons of at least ten years.
- U.S. nominal equity return expectations average about 5.12% across all horizons and 6.16% at horizons of at least ten years.
- Allocation funds average about 34.36% U.S. equity, 18.47% non-U.S. equity, 38.29% bonds, 4.37% cash, and 4.43% other assets.
- Figure A1 and Table A1 show that the dataset contains many providers, but the usable sample is concentrated among large managers with repeated forecasts and matched funds.

Economic message: The data combine professional expectations with actual allocation portfolios, making it possible to study both beliefs and demand.

Table 1
Summary statistics

	<i>N</i>	Mean	Median	Min	Max	Standard deviations across		
						Manager/fund	Horizon	Time
A. Manager equity expectations								
U.S. equity premium (over yield)								
All horizons	360	3.02	3.79	−6.50	11.52	1.75	0.88	1.01
<10-year horizon	181	2.33	3.40	−6.50	11.52			
≥10-year horizon	179	3.72	4.02	−1.25	7.32			
U.S. equity return (nominal level)								
All horizons	383	5.12	5.69	−5.10	11.80	1.63	0.88	0.70
<10-year horizon	181	3.96	5.08	−5.10	11.80			
≥10-year horizon	202	6.16	6.53	−0.10	9.30			
U.S. equity premium (over cash)								
All horizons	343	3.11	3.90	−6.50	11.50	1.70	0.83	0.91
<10-year horizon	164	2.18	3.21	−6.50	11.50			
≥10-year horizon	179	3.97	4.40	−0.70	6.25			
DM equity premium (over yield)								
All horizons	234	3.73	3.81	−1.50	9.20	1.09	0.73	1.15
<10-year horizon	111	3.06	3.20	−1.50	9.20			
≥10-year horizon	123	4.33	4.38	0.15	8.71			
EM equity premium (over yield)								
All horizons	318	5.27	5.50	−1.55	13.22	1.22	0.94	1.24
<10-year horizon	168	5.11	5.27	−1.55	13.22			
≥10-year horizon	150	5.46	5.66	1.45	13.02			
B. Fund shares								
U.S. equity	3139	34.36	32.47	−16.22	99.73	17.39		4.99
Non-U.S. equity	3139	18.47	16.41	−22.95	99.49	10.56		4.00
Bonds	3139	38.29	34.61	0.00	266.49	17.93		7.49
Cash	3139	4.37	4.67	−290.04	100.00	9.93		7.58
Other assets	3139	4.43	1.30	−2.80	77.28	5.02		2.87

The table shows number of observations and summary statistics (mean, median, minimum, and maximum) for asset managers' expectations of U.S. equity premiums (over a matched yield), U.S. equity returns (nominal level), U.S. equity returns over the subjective returns on cash over the same horizon, developed markets equity premiums (DM, over a matched yield), and emerging markets equity premiums (EM, over a matched yield) in panel A, as well as fund shares invested in U.S. equities, non-U.S. equities, bonds, cash, and other assets for asset managers' allocation funds in panel B. The table also shows in panel A average standard deviations across one dimension (manager, horizon, or time) while fixing the other two dimensions; for example, to compute the average standard deviation of equity premium expectations across managers, we fix a horizon and a quarter (time), compute the standard deviation across managers, repeat the same process for all horizon-quarter pairs, and then average across all horizon-quarter pairs. The corresponding standard deviations for fund shares in panel B are only across fund or over time. The manager equity expectations and fund shares are expressed in % per

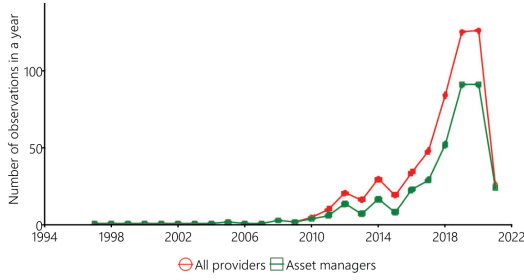


Figure A1
Number of observations per year
The figure shows the number of return forecasts per year for the sample of all providers of return forecasts (red circles) and sample of asset managers that we can match to portfolio data (green squares). The sample period ends in April 2021.

Table A1
List of asset managers

Asset manager	Number of observations				
	(1)	(2)	(3)	(4)	(5)
Amundi	37	13	152	0	2.89
AQR	9	9	71	47	0.19
BlackRock	78	11	297	205	53.41
BMO	3	3	60	60	0.69
BNY Mellon	5	5	68	68	4.45
Columbia Threadneedle	2	2	106	106	25.45
DWS	8	8	148	0	4.29
Franklin Templeton	5	5	214	49	101.04
GMO	47	47	276	276	15.92
Graystone Consulting / Morgan Stanley	7	5	21	9	0.55
Invesco	10	10	202	202	26.99
J.P. Morgan	24	24	558	558	39.48
Morningstar	16	16	50	50	0.50
Northern Trust	10	10	16	0	0.13
PIMCO	3	3	19	19	23.66
Pioneer Investments	6	2	48	0	0.00
State Street	65	19	82	68	0.21
T. Rowe Price	3	3	48	0	69.48
UBS	3	3	12	12	0.53
Vanguard	6	6	160	0	345.66
Voya	6	6	372	372	18.48
Wells Fargo Investment Institute	3	3	141	141	9.46
Total	356	213	3,121	2,242	743.39

The table lists the asset managers in the sample and decomposes the number of observations in key regressions by asset manager. Column 1 refers to specification (1) in Table 2. Column 2 refers to specification (2) in Table 2. Column 3 refers to specification (2) in panel A of Table 5. Column 4 refers to specification (1) in panel A of Table 5. Column 5 shows the 2021 assets under management (AUM) for funds in the sample in column 3 in billions of US\$. BMO refers to Bank of Montreal Global Asset Management, GMO to Grantham, Mayo, & van Otterloo, and PIMCO to Pacific Investment Management Company.

A.3 List of Asset Managers and Sample Composition

Figure A1 shows the number of return forecasts for the sample of all providers and for the sample of

In column 2, the sample is restricted to one equity premium forecast per asset manager per date. By comparing columns 1 and 2, it is apparent which managers provide a term structure of expectations.

(a) Appendix Figure A1: observations over time.

(b) Appendix Table A1: asset managers.

5.2 Table 2, Figure 1, and Figure 2: equity premium expectations and CAPE

Read:

- Table 2 shows that log CAPE is negatively related to subjective equity premium expectations.
- The coefficient on $\log(\text{CAPE})$ is about -5.94 for all-horizon U.S. equity premium expectations over yield and -6.53 for forecasts closest to ten years.
- With controls for past returns and the risk-free rate, the coefficient remains negative and statistically significant.
- Figure 1 shows that asset managers' U.S. equity premium forecasts fall when CAPE rises and rise when CAPE falls.
- Figure 2 shows that the slope across horizons differs at low and high CAPE: low valuations imply higher long-horizon equity premium forecasts.

Economic message: Asset managers' equity premium expectations are countercyclical, aligning with valuation-based asset-pricing logic rather than retail-style extrapolation.

Table 2
Subjective equity return expectations and CAPE

	Equity premium (over yield)		Equity return (nominal level)	Equity premium (over cash)
	All horizons (1)	Closest to 10 years (2)		
A. CAPE			(3)	(4)
$\ln(\text{CAPE})$	-5.942*** (2.052)	-6.526** (2.411)	-4.831* (2.377)	-5.372* (2.639)
N	356	213	379	338
Adjusted R^2	.776	.864	.810	.780
Manager $\times$ Horizon FE	Yes	Yes	Yes	Yes
B. CAPE, past return, and risk-free rate				
$\ln(\text{CAPE})$	-5.252** (2.502)	-6.102** (2.756)	-5.252** (2.502)	-5.671* (2.911)
Past 12-month return	-0.001 (0.008)	-0.007 (0.007)	-0.001 (0.008)	0.012 (0.010)
Risk-free rate	-0.692*** (0.185)	-0.489*** (0.140)	0.308 (0.185)	-0.409* (0.234)
N	356	213	356	315
Adjusted R^2	.824	.885	.812	.796
Manager $\times$ Horizon FE	Yes	Yes	Yes	Yes

The table shows panel regressions of asset managers' U.S. equity return expectations on the log of the cyclically adjusted price-earnings ratio (CAPE). Panel A shows regressions with the CAPE only; panel B shows regressions with the CAPE, the past 12-month return of the S&P 500 index, and the matched yield as the risk-free rate. Specifications (1) and (2) are for equity premiums over yield (nominal equity forecast minus a matched nominal yield), specification (3) for the nominal level of equity returns, and specification (4) for equity premiums over cash (nominal equity forecast minus nominal cash forecast over the same horizon). Specification (1) includes equity premium expectations for all horizons; specification (2) includes, for a given date, only one equity

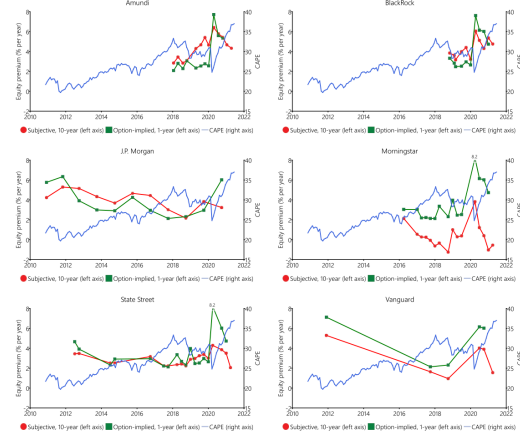


Figure 1
Equity premium expectations
The figure shows 10-year U.S. equity premium expectations for six asset managers (red-filled circles, left axis) and, matched by date, Martin's (2017) 1-year option implied lower bound of the equity premium (green-filled squares, left axis). One observation (8.2% per year for the option-implied equity premium) is outside the plotted ranges. The six asset managers are Amundi, BlackRock, J.P. Morgan, Morningstar, State Street, and Vanguard. These managers (a) have the most data points available for a 10-year horizon and (b) provide both equity return and cash return forecasts. J.P. Morgan provides forecasts with a horizon interval of 10–15 years. The figure

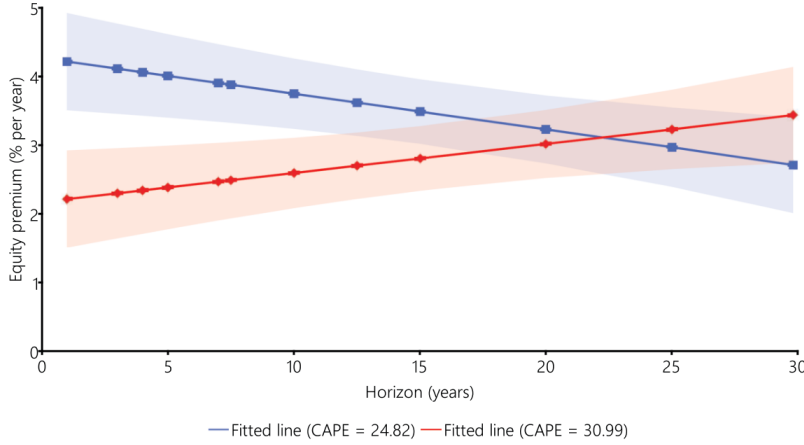


Figure 2
Fitted U.S. equity premium expectations against forecast horizons

The figure shows fitted lines of U.S. equity premium expectations over horizons based on estimates in specification (3) of Table F1 in the Internet Appendix. The blue solid line with filled squares and the red solid line with filled diamonds are conditional on a cyclically adjusted price-earnings ratio (CAPE) of 24.82 (in March 2020) and 30.99 (in January 2020), respectively. The shaded areas represent two-standard-deviation confidence bands.

5.3 Table 3: subjective and objective equity premium expectations

Read:

- The valuation-based equity premium coefficient is 1.074 and statistically significant.
- The regression-based equity premium coefficient is 1.077 and statistically significant.
- Manager-by-horizon fixed effects are included.

Economic message: Professional expectations covary positively with objective expected-return measures, reinforcing the view that these forecasts are valuation-based.

Table 3
Subjective and objective equity return expectations

	Equity premium	
	(1)	(2)
Valuation-based equity premium	1.074*** (0.222)	
Regression-based equity premium		1.077** (0.398)
N	207	213
Adjusted R^2	.856	.864
Manager $\times$ Horizon FE	Yes	Yes

The table shows panel regressions of asset managers' U.S. equity premium expectations (nominal equity forecast minus a matched nominal yield) on measures of objective equity premium expectations. Specification (1) uses a valuation-based measure of objective equity premium expectations, constructed as $\mu = \ln(1 + 1/\text{CAPE}) - \ln(1 + y)$ where y is the real 10-year Treasury yield. Specification (2) uses fitted values from specification (1) in Table C1 in the Internet Appendix. The specification in the Internet Appendix regresses realized 10-year excess returns on the log CAPE using data from 1871 to 2021. The sample includes, for a given date, only one equity premium expectation per asset manager (the one closest to a horizon of 10 years). All specifications include a manager-times-horizon fixed effect, but the fixed effect coefficient estimates are not reported. Standard errors (in parentheses) are clustered by year-month and manager. N refers to the total number of observations.

5.4 Figure 3 and Table 4: subjective term premiums

Read:

- Figure 3 shows subjective term premium expectations for six asset managers together with ACM and Kim–Wright model estimates.
- Table 4 shows that subjective term premium expectations are positively related to model-based objective term premiums.
- The coefficient is 0.559 for the ACM-based term premium and 0.886 for the Kim–Wright-based term premium.

Economic message: The pattern is not limited to equities. Asset managers' bond-market expectations also move with standard objective term-premium models.

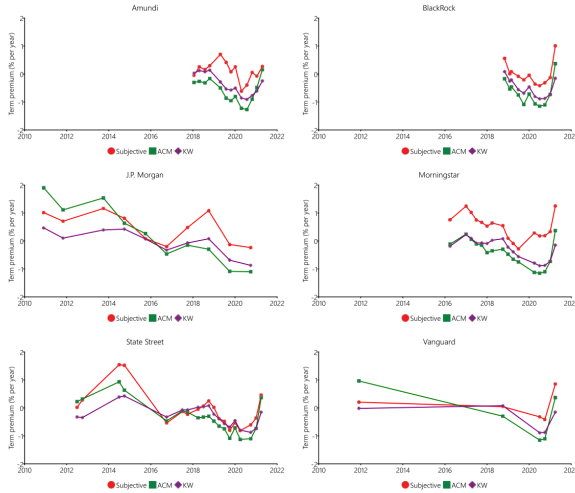


Figure 3
Term premium expectations

The figure shows 10-year term premiums for six asset managers (red-filled circles) and, matched by date, term premiums from the Adrian, Crump, and Moench (2013) (ACM; green-filled squares) and Kim and Wright (2005) (KW; purple-filled diamonds) models. The six asset managers are Amundi, BlackRock, J.P. Morgan, Morningstar, State Street, and Vanguard. These managers have (a) the most data points available for a 10-year horizon, and (b) provide both equity return and cash return forecasts. J.P. Morgan provides forecasts with a

(a) Figure 3: term premium expectations.

Table 4
Subjective and objective term premiums

	Term premium	
	(1)	(2)
ACM-based term premium	0.559*** (0.082)	
KW-based term premium		0.886*** (0.202)
N	239	239
Adjusted R^2	.562	.525
Manager $\times$ Horizon FE	Yes	Yes

The table shows panel regressions of asset managers' U.S. term premium expectations (Treasury yield minus an expected return on cash over the same horizon as the Treasury yield) on measures of objective term premium expectations. Specification (1) uses term premium estimates from the Adrian, Crump, and Moench (2013) model. Specification (2) uses term premium estimates from the Kim and Wright (2005) model. All specifications include a manager-times-horizon fixed effect, but the fixed effect coefficient estimates are not reported. Standard errors (in parentheses) are clustered by year-month and manager. N refers to the total number of observations. * $p < .1$; ** $p < .05$; *** $p < .01$ (for the null hypothesis of a zero coefficient).

until 2020. We find that this pattern is mirrored in asset managers' subjective expectations.

Table 4 presents regressions of asset managers' subjective term premiums using all forecast horizons on the horizon-matched Adrian, Crump, and

(b) Table 4: subjective and objective term premiums.

5.5 Table 5, Figure 4, and Table 6: portfolio shares and U.S. equity expectations

Read:

- Table 5 shows that allocation funds managed by firms with higher U.S. equity premium expectations hold more U.S. equities.
- In the level specification, the coefficient on U.S. expectations is 2.051 without controls and 2.287 when controlling for DM and EM expectations.
- In the log specification, the coefficient on U.S. expectations is 9.602 without controls and 10.760 when controlling for DM and EM expectations.
- Figure 4 visualizes the positive relationship using residualized bins.
- Table 6 adds fund fixed effects and the relation remains positive but smaller, consistent with mandates muting portfolio response.

Economic message: Subjective expectations matter for actual portfolios, but the response is constrained by fund mandates and strategic allocation rules.

Table 5
U.S. equity share and subjective equity return expectations

	U.S. equity share		
	(1)	(2)	(3)
A. Level specification			
U.S. expectations	2.051*** (0.298)	3.983*** (0.292)	2.287*** (0.270)
DM expectations		-4.615*** (1.189)	
EM expectations		1.818* (0.923)	
<i>N</i>	3,121	2,242	2,242
Adjusted R^2	.074	.134	.093
Year-month FE	Yes	Yes	Yes
B. Log specification			
U.S. expectations	9.602*** (1.533)	15.011*** (2.449)	10.760*** (1.358)
DM expectations		-18.214** (7.118)	
EM expectations		13.379 (8.116)	
<i>N</i>	3,066	2,192	2,192
Adjusted R^2	.084	.117	.097
Year-month FE	Yes	Yes	Yes

Panel A shows regressions of the U.S. equity shares of asset managers' allocation funds on the U.S. market, developed markets (DM), and emerging markets (EM) equity return expectations. Return expectations are equity premiums (nominal equity forecast minus a matched nominal yield). Panel B shows regressions of the log U.S. equity shares of asset managers' allocation funds on the same return expectations. All specifications include a year-month fixed effect. Standard errors (in parentheses) are clustered by year-month and manager. *N* refers to the total number of observations. * $p < .1$; ** $p < .05$; *** $p < .01$ (for the null hypothesis of a zero coefficient).

(a) Table 5: U.S. equity share and expectations.

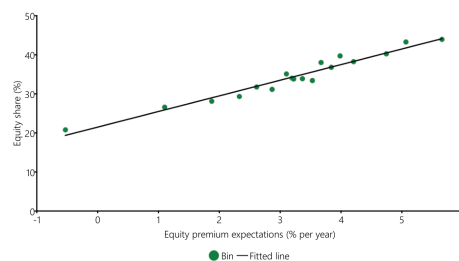


Figure 4
U.S. equity shares and equity return expectations
The figure shows a conditional binscatter plot of U.S. equity shares (the fraction of U.S. equity in a fund's portfolio) and asset managers' U.S. equity premium expectations, conditional on year-month fixed effects and developed as well as emerging market equity premium expectations (the controls). Before binning and plotting, we compute residuals from a regression of U.S. equity shares and U.S. equity premium expectations on the fixed effects and the controls. We add back the sample means of the U.S. equity share and the U.S. equity premium expectation. We then group the residualized U.S. equity shares and U.S. equity premium expectations into 18 equal-sized bins, compute the mean within each bin, and create a scatterplot of the resultant data points.

(b) Figure 4: binned relationship.

Table 6
U.S. equity share and subjective equity return expectations (fund fixed effects)

	U.S. equity share		
	(1)	(2)	(3)
A. Level specification			
U.S. expectations	1.070** (0.448)	2.081*** (0.607)	1.738*** (0.337)
DM expectations		-0.751 (0.554)	
EM expectations		0.309 (0.422)	
<i>N</i>	3,118	2,240	2,240
Adjusted <i>R</i> ²	.888	.893	.893
Fund FE	Yes	Yes	Yes
Year-month FE	Yes	Yes	Yes
B. Log specification			
U.S. expectations	9.806** (3.507)	11.872** (4.355)	16.724*** (3.224)
DM expectations		-1.083 (7.512)	
EM expectations		6.081 (5.735)	
<i>N</i>	3,063	2,190	2,190
Adjusted <i>R</i> ²	.536	.542	.541
Fund FE	Yes	Yes	Yes
Year-month FE	Yes	Yes	Yes

Panel A shows regressions of the U.S. equity shares of asset managers' allocation funds on the U.S. market, developed markets (DM), and emerging markets (EM) equity return expectations. Return expectations are equity premiums (nominal equity forecast minus a matched nominal yield). Panel B shows regressions of the log U.S. equity shares of asset managers' allocation funds on the same return expectations. All specifications include a fund fixed effect and a year-month fixed effect. Standard errors (in parentheses) are clustered by year-month and

5.6 Tables 7–9: tactical funds, aggregate equity shares, and bonds

Read:

- Table 7 shows that tactical allocation funds have stronger sensitivity to U.S. equity premium expectations, especially in log specifications.
- Table 8 shows that aggregate equity shares increase with aggregate equity premium expectations; the coefficient is 1.424 in levels and 4.545 in logs.
- Tactical-fund interactions are positive in Table 8, but imprecisely estimated.
- Table 9 shows little systematic effect on overall bond shares, but term premium expectations increase fixed-income duration and maturity.

Economic message: Expectations show up most clearly where the portfolio margin is natural: equity premium expectations affect equity shares, while term premium expectations affect duration and maturity.

Table 7
U.S. equity share and subjective equity return expectations (tactical funds)

	U.S. equity share	
	(1)	(2)
A. Level specification		
U.S. expectations	3.867*** (0.206)	1.686** (0.611)
Tactical fund	−0.488*** (0.071)	−0.252*** (0.067)
U.S. expectations × Tactical fund	5.425** (1.608)	6.144*** (1.269)
DM expectations	−3.282*** (0.404)	−0.430 (0.542)
EM expectations	0.936* (0.460)	0.138 (0.412)
<i>N</i>	2,242	2,240
Adjusted <i>R</i> ²	.211	.896
Fund FE	No	Yes
Year-month FE	Yes	Yes
B. Log specification		
U.S. expectations	14.614*** (2.082)	9.396* (4.323)
Tactical fund	−2.310*** (0.447)	−1.693*** (0.317)
U.S. expectations × Tactical fund	29.191** (9.318)	36.300*** (6.445)
DM expectations	−13.524** (5.281)	0.760 (8.038)
EM expectations	10.151 (7.434)	5.241 (5.945)
<i>N</i>	2,192	2,190
Adjusted <i>R</i> ²	.160	.546
Fund FE	No	Yes
Year-month FE	Yes	Yes

Panel A shows regressions of U.S. equity shares of asset managers' allocation funds on U.S., developed markets (DM), and emerging markets (EM) equity return expectations, allowing for specific sensitivity to tactical funds. Return expectations are equity premiums (nominal equity forecast minus a matched nominal yield). The variable "Tactical fund" is a dummy variable that takes a value of one if the U.S. equity share is for a tactical allocation fund. Panel B shows regressions of the log U.S. equity shares of asset managers' allocation funds on the same variables. The specifications allow for year-month fixed effects. Specifications (2) also include a fund fixed effect.

Table 8
Aggregate equity share and subjective equity return expectations

	Aggregate equity share	
	(1)	(2)
A. Level specification		
Equity expectations	1.424*** (0.334)	1.343*** (0.342)
Tactical fund		−0.155 (0.169)
Equity expectations × Tactical fund		2.837 (3.989)
<i>N</i>	2,240	2,240
Adjusted <i>R</i> ²	.897	.898
Fund FE	Yes	Yes
Year-month FE	Yes	Yes
B. Log specification		
Equity expectations	4.545** (1.659)	4.119** (1.599)
Tactical fund		−0.701 (0.610)
Equity expectations × Tactical fund		14.148 (14.720)
<i>N</i>	2,209	2,209
Adjusted <i>R</i> ²	.767	.769
Fund FE	Yes	Yes
Year-month FE	Yes	Yes

Panel A shows regressions of equity shares of asset managers' allocation funds on global equity premium expectations. The global equity premium expectation at a given point in time is constructed as a weighted average of U.S. equity premium, developed markets equity premium expectations, and emerging markets equity premium expectations with market-capitalization weights from the MSCI ACWI index. The variable "Tactical fund" is a dummy variable that takes a value of one if the equity share is for a tactical allocation fund. Panel B shows regressions of the log equity shares of asset managers' allocation funds on global equity premium expectations.

(a) Table 8: aggregate equity share.

Table 9
Bond share, duration, maturity, and subjective term premium expectations

	Bond share		Duration		Maturity	
	(1)	(2)	(3)	(4)	(5)	(6)
A. Level specification						
U.S. expectations	0.933 (2.420)	0.066 (0.940)	1.264* (0.694)	0.587** (0.236)	2.358** (1.020)	1.047** (0.418)
<i>N</i>	2,490	2,482	504	480	440	418
Adjusted <i>R</i> ²	.022	.828	.151	.845	.176	.871
Fund FE	No	Yes	No	Yes	No	Yes
Year-month FE	Yes	Yes	Yes	Yes	Yes	Yes
B. Log specification						
U.S. expectations	−14.075 (21.339)	4.870 (3.332)	0.360 (0.300)	0.071* (0.039)	0.412 (0.299)	0.076* (0.036)
<i>N</i>	2,475	2,468	503	479	440	418
Adjusted <i>R</i> ²	−.009	.870	.053	.896	.069	.959
Fund FE	No	Yes	No	Yes	No	Yes
Year-month FE	Yes	Yes	Yes	Yes	Yes	Yes

Panel A shows regressions of the bond shares, asset-weighted fixed income durations, and asset-weighted fixed income maturities of asset managers' allocation funds on U.S. term premium expectations (Treasury yield minus an expected return on cash over the same horizon as the Treasury yield). Panel B shows regressions of the log of the variables in panel A on the same term premium expectations. The specifications allow for a year-month fixed effect. Specifications (2), (4), and (6) also include a fund fixed effect. Standard errors (in parentheses) are clustered by year-month and manager. *N* refers to the total number of observations. * *p* < .1; ** *p* < .05; *** *p* < .01 (for the null hypothesis of a zero coefficient).

regressions of bond portfolio shares on U.S. term premium expectations. Specification (1) includes year-month fixed effects and is analogous to the specification in Table 8. Specifications (2) through (6) also include a fund fixed effect.

(b) Table 9: bond shares, duration, and maturity.

5.7 Figure 5 and Tables 10–11: CFOs and professional forecasters

Read:

- Figure 5 compares CFO and professional forecaster expectations with CAPE.
- CFOs' one-year equity premium expectations are countercyclical in Table 10, but their nominal equity return expectations are procyclical.

- CFOs' ten-year equity premium expectations are not significantly related to CAPE.
- Professional forecasters' one-year equity premium expectations are strongly countercyclical in Table 11, but their ten-year expectations are not significantly related to CAPE.

Economic message: The institutional asset-manager results are distinct from prior evidence on retail expectations and also differ from CFO and professional-forecaster evidence at some horizons.

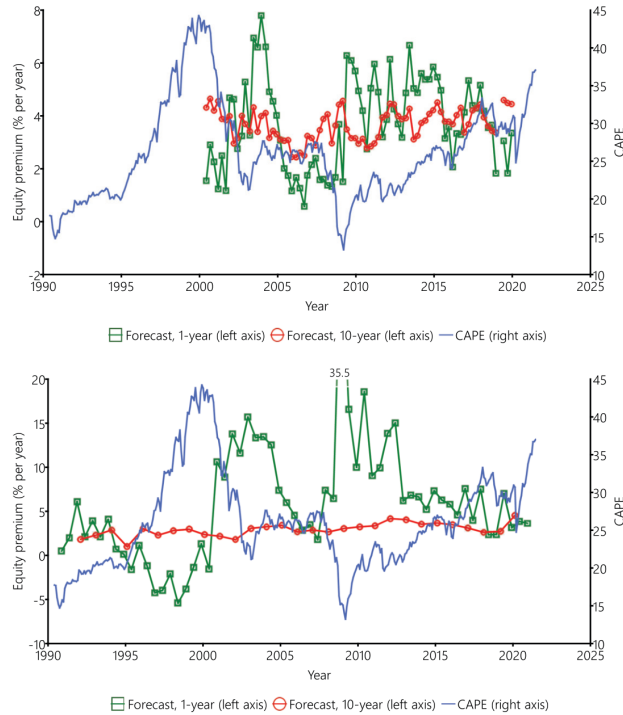


Figure 5

CFOs' and professional forecasters' expectations and CAPE

The top panel shows chief financial officers' (CFOs') average 1- and 10-year U.S. equity premium expectations (green squares and red circles; left axis) and Shiller's cyclically adjusted price-earnings ratio (CAPE; solid blue line; right axis). The bottom panel shows the average 1- and 10-year U.S. equity premium expectations of professional forecasters (green squares and red circles; left axis) and the CAPE (solid blue line; right axis). The sample period for CFOs' expectations is from Q2:2000 to Q4:2018. The sample period for professional

Table 10
CFOs' expectations and CAPE

	Equity premium (1)	Equity return (2)
A. 1-year horizon		
ln(CAPE)	-2.089** (0.976)	3.468*** (0.875)
Constant	10.525*** (3.169)	-5.603* (5.891)
N	75	75
Adjusted R ²	.208	.277
B. 10-year horizon		
ln(CAPE)	0.506 (0.548)	3.014** (1.488)
Constant	2.006 (1.783)	-2.658 (4.766)
N	75	75
Adjusted R ²	.043	.017

The table shows time-series regressions of chief financial officers' (CFOs') U.S. equity return expectations on the log of the cyclically adjusted price-earnings ratio (CAPE). Panel A shows expectations over 1-year horizons, and panel B shows expectations over 10-year horizons. Specification (1) is for equity premiums (equity return minus either a 1- or 10-year yield); specification (2) is for equity returns. Standard errors (in parentheses) are Newey and West (1987) standard errors with four lags. N refers to the total number of observations. *p < .1; **p < .05; ***p < .01 (for the null hypothesis of a zero coefficient).

Table 11
Professional forecasters' expectations and CAPE

	Equity premium (1)	Equity return (2)
A. 1-year horizon		
ln(CAPE)	-13.740*** (4.680)	-11.029*** (4.487)
N	1,318	1,318
Adjusted R ²	.298	.246
Forecaster FE	Yes	Yes
B. 10-year horizon		
ln(CAPE)	-0.248 (0.524)	0.377 (0.579)
N	681	681
Adjusted R ²	.229	.284
Forecaster FE	Yes	Yes

The table shows panel regressions of professional forecasters' U.S. equity return expectations on the log of the cyclically adjusted price-earnings ratio (CAPE). Panel A shows expectations over 1-year horizons, and panel B shows expectations over 10-year horizons. Specification (1) is for equity premiums (equity return minus either a 1- or 10-year yield); specification (2) is for equity returns. All specifications include a forecaster fixed effect, but the fixed effect coefficient estimates are not reported. Standard errors (in parentheses) in panel A are clustered by semiyear and forecaster, and in panel B by year and forecaster. N refers to the total number of observations. *p < .1; **p < .05; ***p < .01 (for the null hypothesis of a zero coefficient).

price-dividend ratio in their case) of 3.40 is close to our estimate of 3.47.

(a) Table 10: CFOs' expectations and CAPE.

(b) Table 11: professional forecasters.

6 Short summary

- The paper collects capital market assumptions from large asset managers and matches them to allocation fund portfolios.
- Asset managers' U.S. equity premium expectations are countercyclical, falling with the CAPE.
- Their expectations comove positively with valuation-based and regression-based objective equity premium forecasts.
- Their subjective term premiums comove with ACM and Kim–Wright term premium measures.
- Allocation funds managed by firms with higher U.S. equity premium expectations hold more U.S. equities.
- Fund fixed effects reduce the coefficient, suggesting mandates and target allocations mute the pass-through from beliefs to portfolios.
- Tactical funds are more responsive to expectations.
- Term premium expectations map more naturally into duration and maturity than into overall bond shares.
- The portfolio response is statistically meaningful but smaller than the standard frictionless mean-variance benchmark.

7 Conclusion

7.1 Core conclusion

Dahlquist and Ibert show that large asset managers' equity premium expectations are countercyclical and economically meaningful. Professional investors' beliefs are high when valuations are low, and their portfolios tilt more toward U.S. equities when their expected U.S. equity premium is higher.

7.2 Economic intuition

Large asset managers appear to use valuation-based expected return logic rather than simple extrapolation. However, delegated asset management is not a frictionless mean-variance exercise. Mandates, benchmark rules, strategic target allocations, and risk budgets dampen how much expectations can move portfolio weights.

7.3 Takeaway

The paper bridges beliefs and portfolios for institutional investors. It shows that subjective expectations are not just survey artifacts: they are connected to asset pricing variables and to portfolio demand. But the magnitude of the portfolio response is constrained by the institutional structure of asset management.